

PHASE 3

PROJECT DESIGN

3.1 SETUP ENVIRONMENT & DEVOPS WORKFLOW

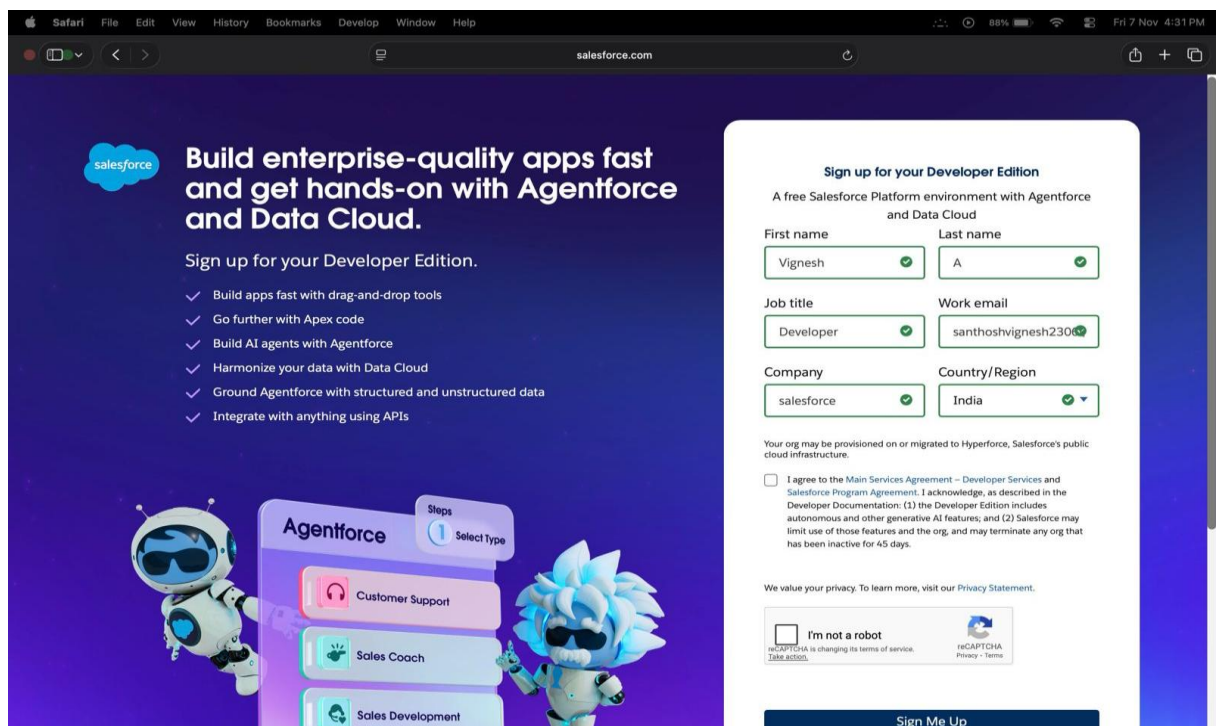
MILESTONE 1- SALESFORCE ACCOUNT

We established a robust Salesforce development environment using **Developer Org Strategy**- Created Developer Org sandboxes for development and testing.

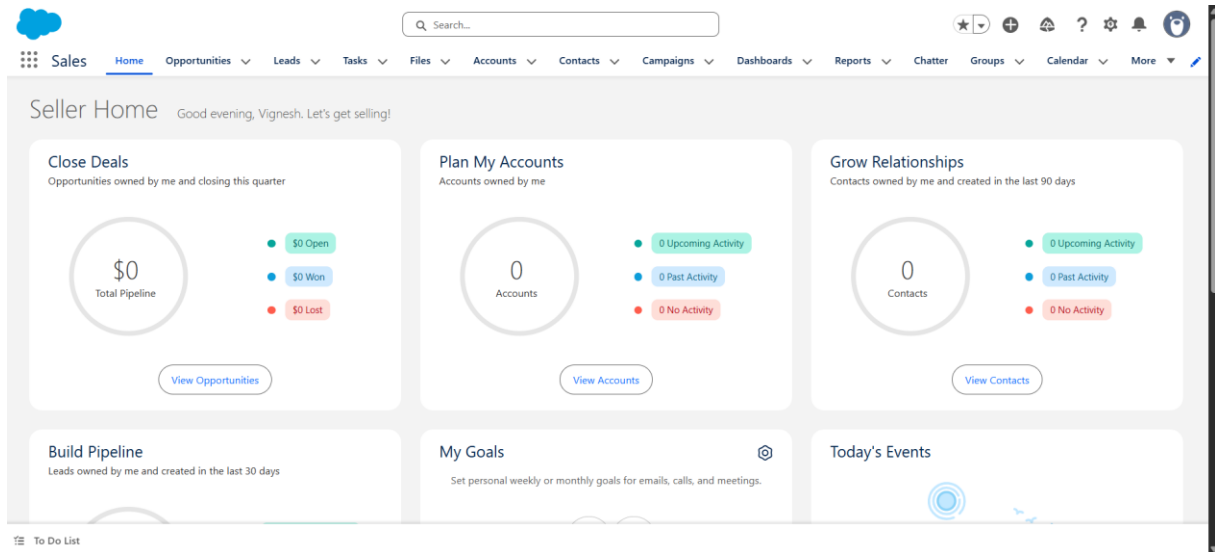
We can create a Developer Org using this official Salesforce link:
<https://developer.salesforce.com/signup>

- **Source Control:** Used GitHub to manage code versions, collaborate with the team, and track all changes efficiently.
- **Deployment Tools:** Utilized Change Sets and Salesforce DX (SFDX CLI) to deploy components between Sandbox and Production environments.
- **Testing Environment:** All features were validated in the Sandbox to ensure quality and stability before live deployment.

This setup supports faster development, safer releases, and better tracking of changes during the entire project lifecycle.



Now, enter your credentials to login into the Salesforce Developer Org.



3.2 CUSTOMIZATION OF OBJECTS

Salesforce's declarative tools are used to customize the system:

MILESTONE 2 - OBJECTS

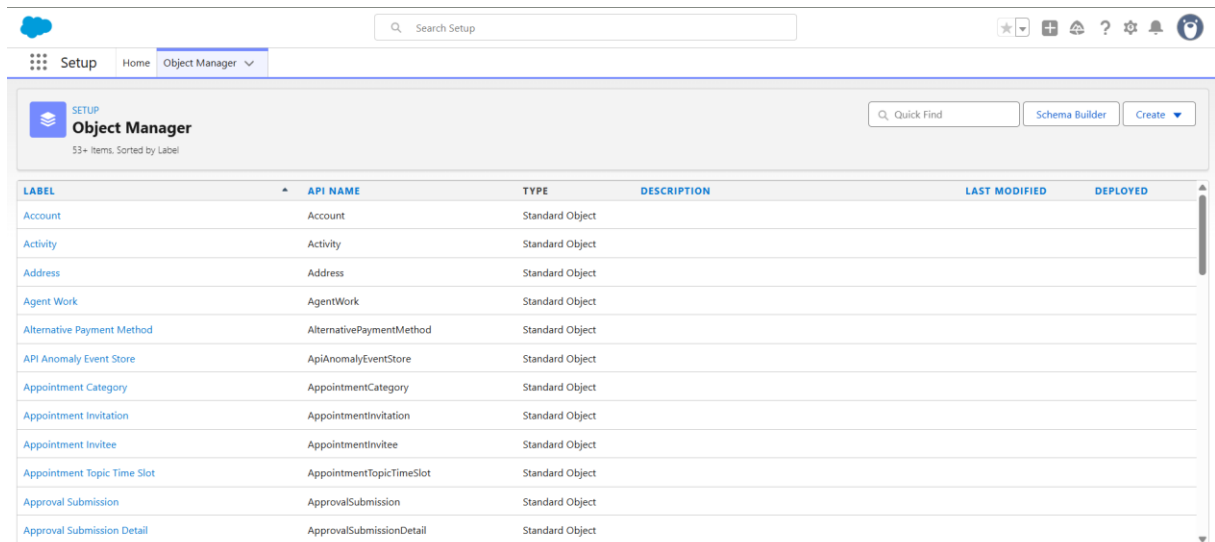
To model lease data, the following custom objects are created:

- **Property**
- **Tenant**
- **Lease**
- **Payment**

Salesforce's declarative tools were used to create custom objects tailored to the Lease Management System.

These objects include **Property, Tenant, Lease, and Payment**, each representing core business data.

They enable structured storage and management of lease-related records. Custom fields, relationships, and validations were added to meet specific project needs.



The screenshot shows the Salesforce Object Manager interface. At the top, there's a navigation bar with 'Setup', 'Home', and 'Object Manager' tabs. Below this, the 'Object Manager' section is active, displaying a list of 53+ items sorted by label. The table below lists several standard objects.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Account	Account	Standard Object			
Activity	Activity	Standard Object			
Address	Address	Standard Object			
Agent Work	AgentWork	Standard Object			
Alternative Payment Method	AlternativePaymentMethod	Standard Object			
API Anomaly Event Store	ApiAnomalyEventStore	Standard Object			
Appointment Category	AppointmentCategory	Standard Object			
Appointment Invitation	AppointmentInvitation	Standard Object			
Appointment Invitee	AppointmentInvitee	Standard Object			
Appointment Topic Time Slot	AppointmentTopicTimeSlot	Standard Object			
Approval Submission	ApprovalSubmission	Standard Object			
Approval Submission Detail	ApprovalSubmissionDetail	Standard Object			

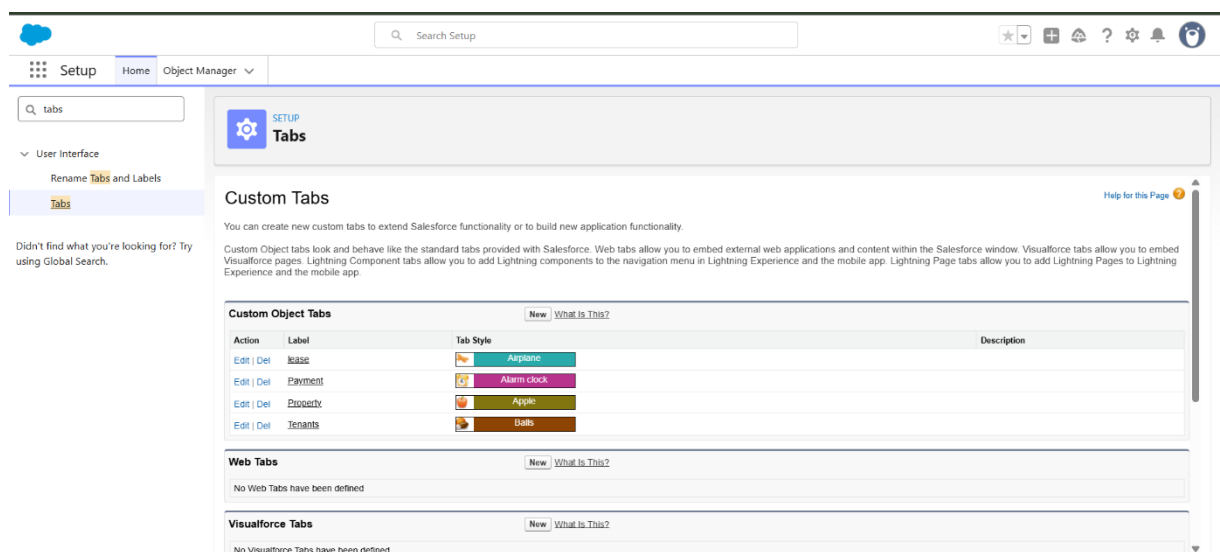
3.3 CUSTOMIZATION OF TABS

MILESTONE 3 - TABS

Lightning Record Pages were customized for each object to improve how data is displayed to users:

- **Use of Tabs:** Tabs were used to organize related information clearly within the record page, ensuring a clean and structured layout.

This setup creates a simple, role-friendly interface, making navigation easier and enhancing the overall user experience.



The screenshot shows the Salesforce Setup interface for 'Tabs'. The left sidebar has a search bar with 'tabs' and a list of categories including 'User Interface' and 'Rename Tabs and Labels'. The main content area is titled 'Custom Tabs' and includes a description: 'You can create new custom tabs to extend Salesforce functionality or to build new application functionality.' Below this, there are three sections: 'Custom Object Tabs', 'Web Tabs', and 'Visualforce Tabs'. Each section has a 'New' button and a 'What Is This?' link. The 'Custom Object Tabs' section shows a table with columns for Action, Label, Tab Style, and Description.

Action	Label	Tab Style	Description
Edit Del	Lease	Airplane	
Edit Del	Payment	Alarm clock	
Edit Del	Property	Apple	
Edit Del	Tenants	Balls	

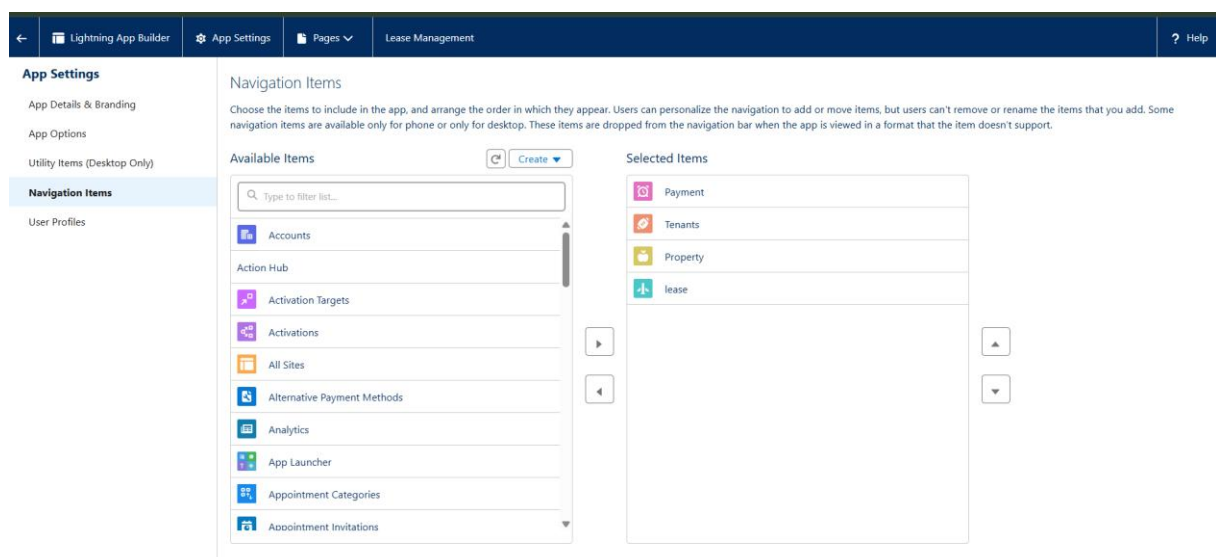
3.4 LIGHTNING APP SETUP THROUGH APP MANAGER

MILESTONE 4: THE LIGHTNING APP

Using App Manager, a custom “**Lease Management**” **Lightning App** is created with a branded name, logo, and navigation items such as:

- Tenants
- Properties
- Payments

This setup ensures quick access to all key objects in one place, improving user workflow.



The next step involves in selecting the system administrator as the default user for the user profiles.

3.5 CUSTOMIZATION OF FIELDS

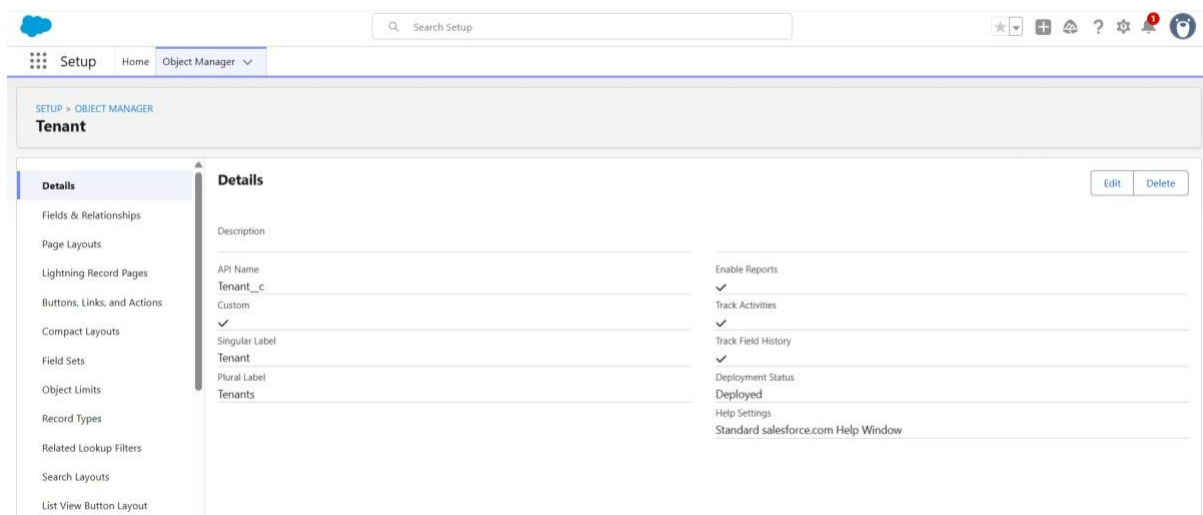
MILESTONE 5 - FIELDS

Each object has custom fields such as:

➤ Tenant:

- Email__c
- Phone__c
- Property__c
- Status__c

➤ Lease:



- End_date__c
- Property__c
- Start_date__c

Setup > Object Manager > lease

Details

Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout

Details

Description

API Name
lease__c

Custom
✓

Singular Label
lease

Plural Label
lease

Enable Reports
✓

Track Activities
✓

Track Field History
✓

Deployment Status
Deployed

Help Settings
Standard salesforce.com Help Window

Edit Delete

➤ Payment For Tenant:

- Amount__c
- Payment_date__c
- Check_for_payment__c
- Property__c
- Tenant__c

Setup > Object Manager > Payment for tenant

Details

Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout

Details

Description

API Name
Payment_for_tenant__c

Custom
✓

Singular Label
Payment for tenant

Plural Label
Payment

Enable Reports
✓

Track Activities
✓

Track Field History
✓

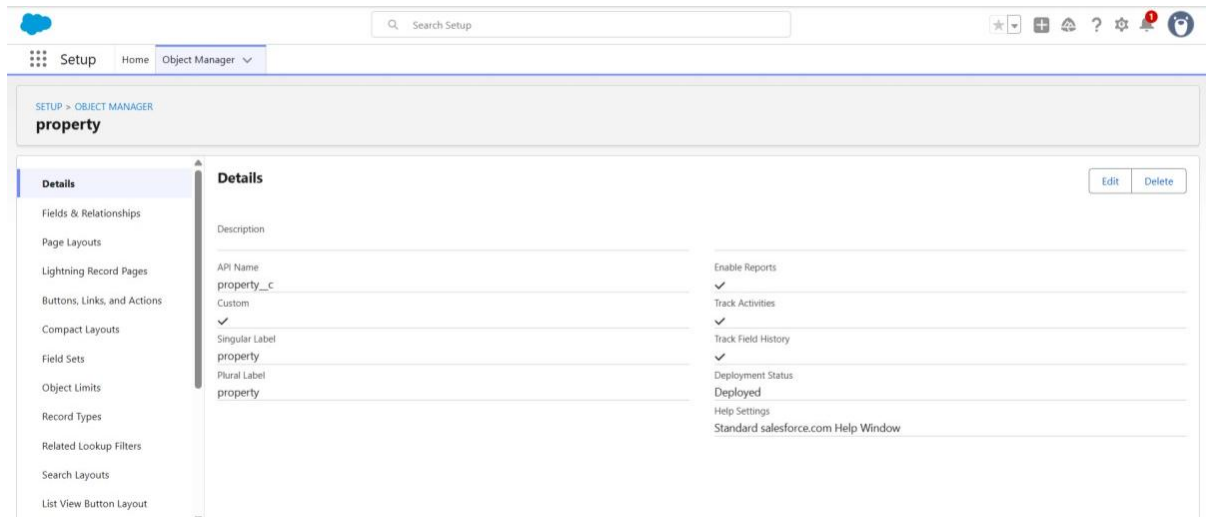
Deployment Status
Deployed

Help Settings
Standard salesforce.com Help Window

Edit Delete

➤ Property:

- Address__c
- Name__c
- Sqft__c
- Type__c



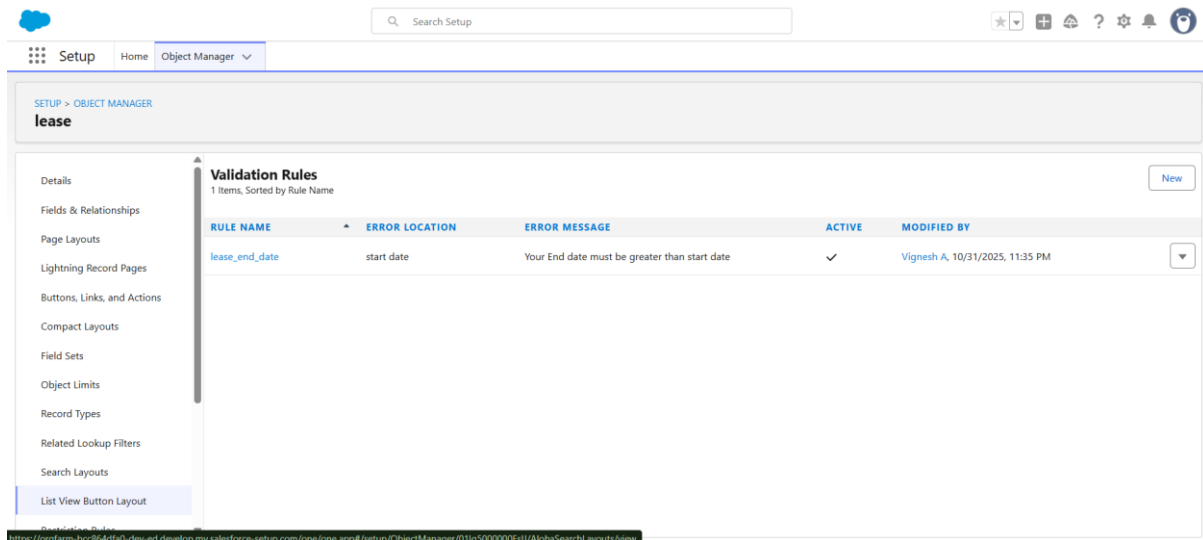
3.6 VALIDATION RULES

MILESTONE 6 - VALIDATION RULE

Validation rules ensure data accuracy.

Examples:

- Lease end date must be after the start date.
- Rent amount must be greater than zero.
- Payment date cannot be in the future (unless marked as scheduled).

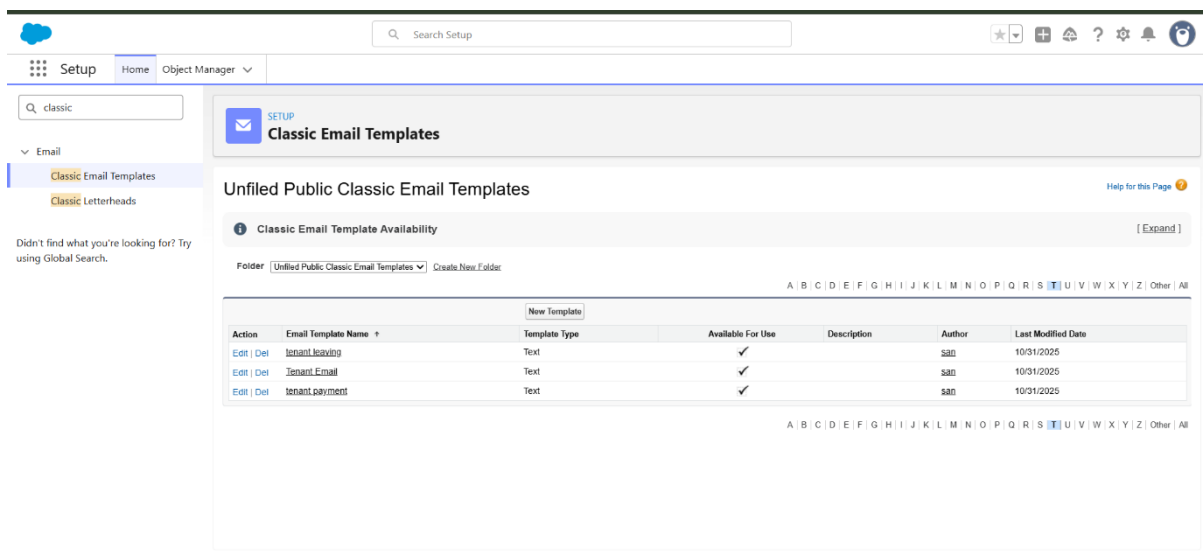


MILESTONE 7: EMAIL TEMPLATES

To ensure timely reminders for rent payments, a scheduled Apex class was developed and tested. The scheduler runs monthly on the 1st day to send payment reminder emails. Testing involved:

- Manually triggering the scheduled job to simulate its execution.
- Verifying emails were dispatched to all tenants with pending payments. Test

documentation includes logs from the scheduler and screenshots of sent emails.



Setup

Home

Object Manager

Q classic

Classic Email Templates

Classic Letterheads

Didn't find what you're looking for? Try using Global Search.

Classic Email Templates

Unfiled Public Classic Email Templates

Classic Email Template Availability

Folder: Unfiled Public Classic Email Templates

Table with 6 columns: Action, Email Template Name, Template Type, Available For Use, Description, Author, Last Modified Date. Rows include Leave approved and Leave rejected.

➤ **Approval Processes:** Used for scenarios like lease renewal approval by a manager.

MILESTONE 8 - APPROVAL PROCESS

Setup

Home

Object Manager

Q appro

Mass Transfer Approval Requests

Approval Settings

Approval Processes

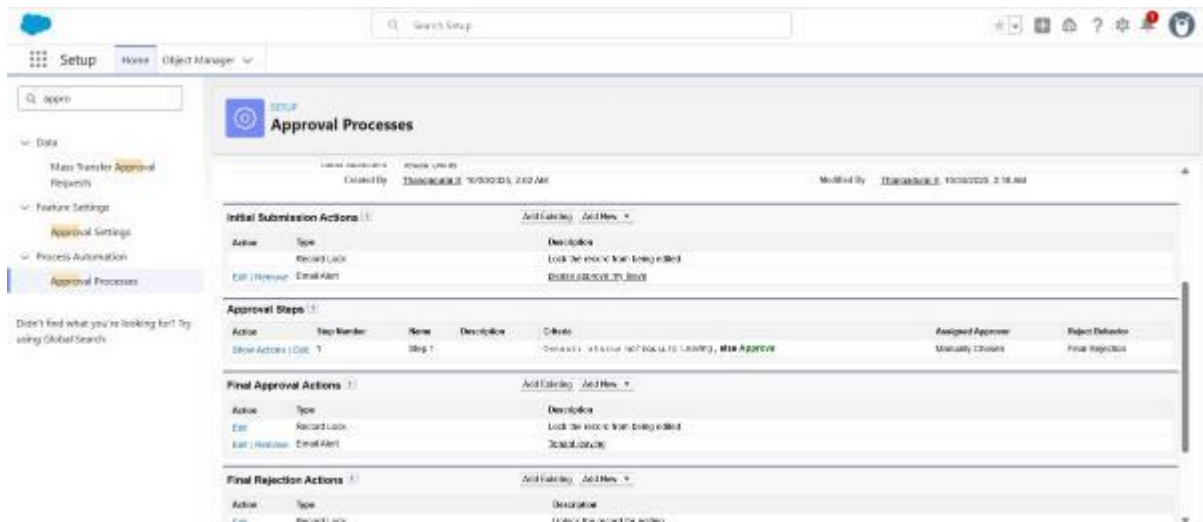
Didn't find what you're looking for? Try using Global Search.

Approval Processes

Manage Approval Processes For: Tenant

Active Approval Processes

Inactive Approval Processes



3.7 APEX TRIGGERS

MILESTONE 9 - APEX TRIGGER

When point-and-click tools are not enough, Apex programming is used to implement custom logic.

Triggers are used to perform actions automatically when a record is inserted, updated, or deleted.

Example:

- On creation of a Lease record, auto-generate a related Payment schedule.
- On Payment update, update Lease status if fully paid.

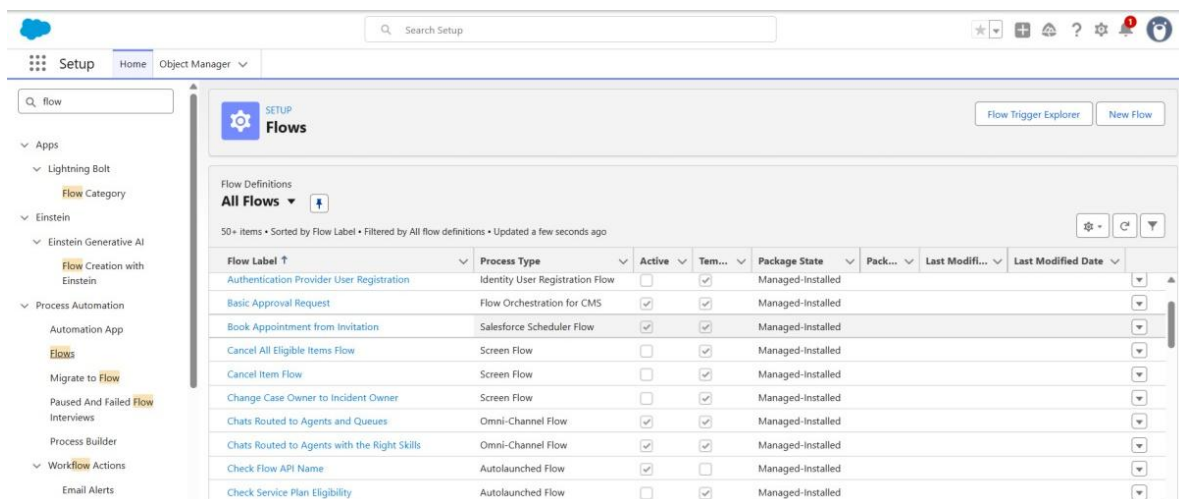
```
File • Edit • Debug • Test • Workspace • Help • < • >
testHandler.apxc
Code Coverage: None • API Version: 65
Go To

1 public class testHandler {
2
3     public static void preventInsert(List<Tenant__c> newList) {
4
5         Set<Id> existingPropertyIds = new Set<Id>();
6
7         for (Tenant__c existingTenant : [SELECT Id, property__c FROM Tenant__c WHERE property__c != null]) {
8
9             existingPropertyIds.add(existingTenant.property__c);
10
11         }
12
13
14         for (Tenant__c newTenant : newList) {
15
16
```

3.8 AUTOMATION TOOLS

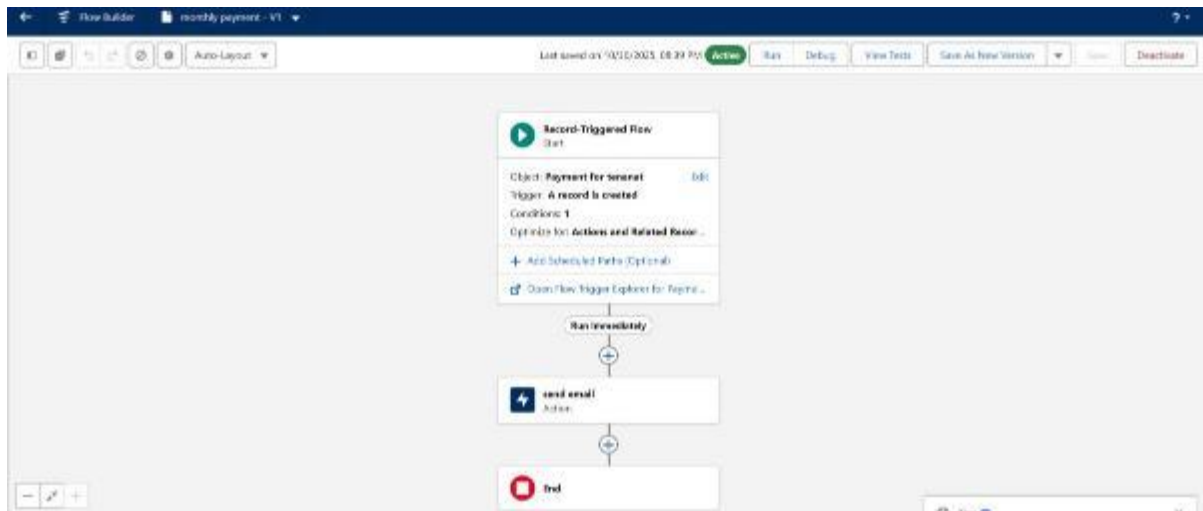
➤ **Flows:** Used to collect tenant input, auto-create payment records, and send reminders.

MILESTONE 10 - FLOWS



The screenshot shows the Salesforce Setup interface, specifically the 'Flows' section. The left sidebar contains navigation links for Setup, Home, Object Manager, and various automation tools like Lightning Bolt, Einstein, and Process Automation. The main content area displays a table of 'Flow Definitions' under the 'All Flows' filter. The table lists various flows such as 'Authentication Provider User Registration', 'Basic Approval Request', and 'Book Appointment from Invitation', along with their process types, active status, and package states.

Flow Label ↑	Process Type	Active	Tem...	Package State	Pack...	Last Modifi...	Last Modified Date
Authentication Provider User Registration	Identity User Registration Flow	☐	☒	Managed-Installed			
Basic Approval Request	Flow Orchestration for CMS	☒	☒	Managed-Installed			
Book Appointment from Invitation	Salesforce Scheduler Flow	☒	☒	Managed-Installed			
Cancel All Eligible Items Flow	Screen Flow	☐	☒	Managed-Installed			
Cancel Item Flow	Screen Flow	☐	☒	Managed-Installed			
Change Case Owner to Incident Owner	Screen Flow	☐	☒	Managed-Installed			
Chats Routed to Agents and Queues	Omni-Channel Flow	☒	☒	Managed-Installed			
Chats Routed to Agents with the Right Skills	Omni-Channel Flow	☒	☒	Managed-Installed			
Check Flow API Name	Autolaunched Flow	☒	☐	Managed-Installed			
Check Service Plan Eligibility	Autolaunched Flow	☐	☒	Managed-Installed			



3.9 APEX CLASSES

These are reusable blocks of code used for custom backend logic.

```
File • Edit • Debug • Test • Workspace • Help • < >
test.apxt testHandler.apxc
Code Coverage: None API Version: 65 Go To
1 trigger test on Tenant_c (before insert)
2
3 {
4
5     if(trigger.isInsert && trigger.isBefore){
6
7         testHandler.preventInsert(trigger.new);
8
9     }
10
11 }
```

Logs Tests Checkpoints Query Editor View State Progress Problems

Name	Line	Problem
------	------	---------

```
File • Edit • Debug • Test • Workspace • Help • < >
test.apxc  testHandler.apxc  MonthlyEmailScheduler.apxc
Code Coverage: None  API Version: 65  Go To

1  global class MonthlyEmailScheduler implements Schedulable {
2
3  global void execute(SchedulableContext sc) {
4
5      Integer currentDay = Date.today().day();
6
7      if (currentDay == 1) {
8
9          sendMonthlyEmails();
10
11      }
12
13  }
14
15
16  public static void sendMonthlyEmails() {
```

Logs Tests Checkpoints Query Editor View State Progress Problems

Name	Line	Problem
------	------	---------

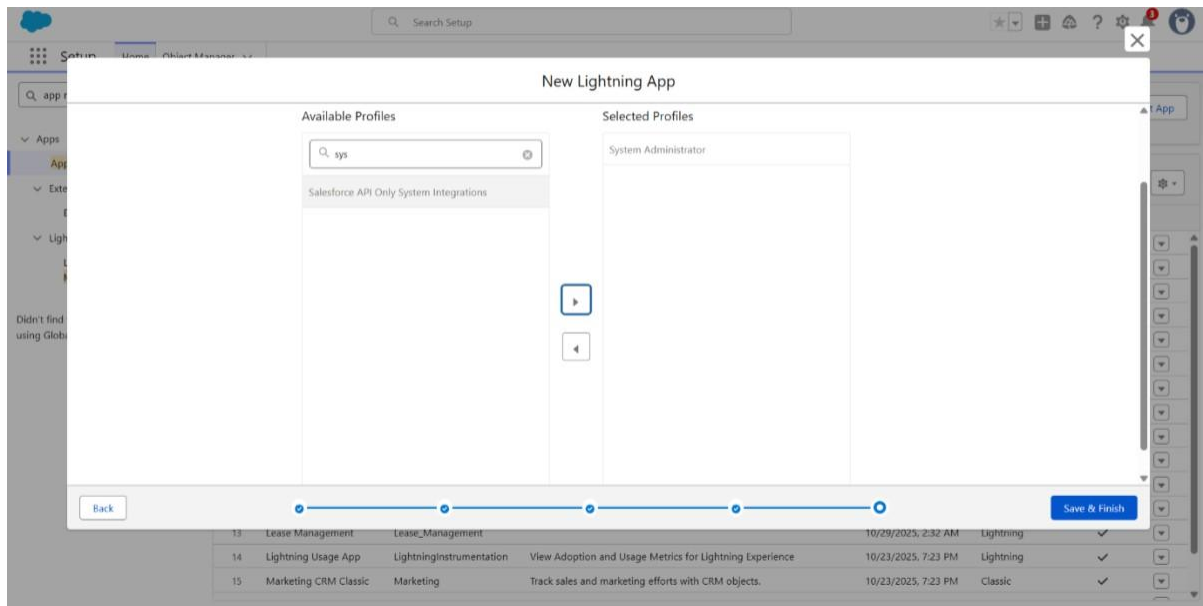
3.10 ASYNCHRONOUS APEX

MILESTONE 11: SCHEDULE CLASS

Used when tasks take longer to run or need to happen in the background:

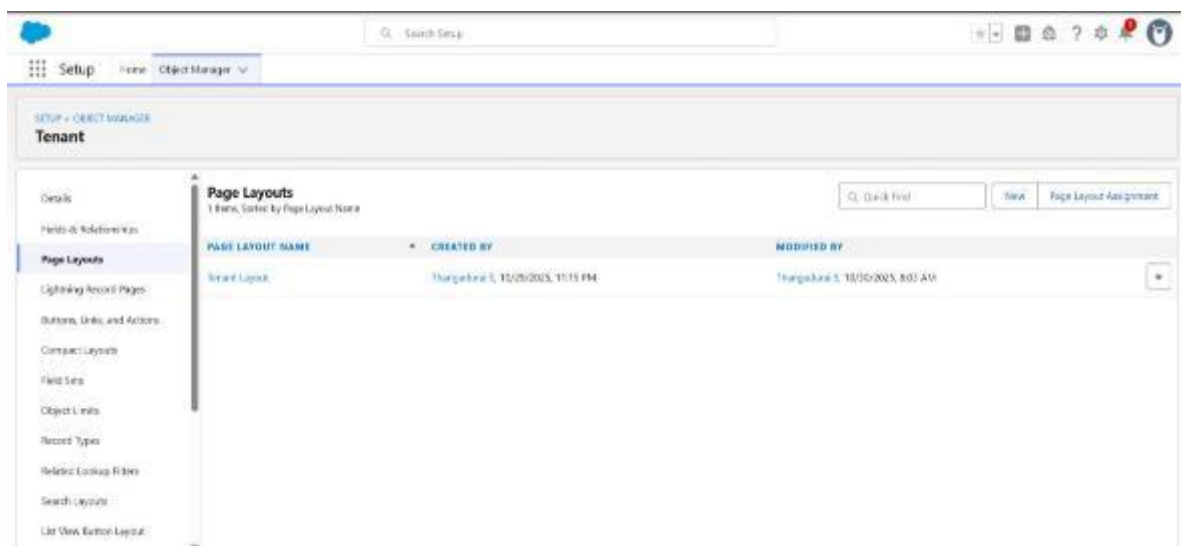
- **Scheduled Apex:** To check daily for upcoming lease expirations.

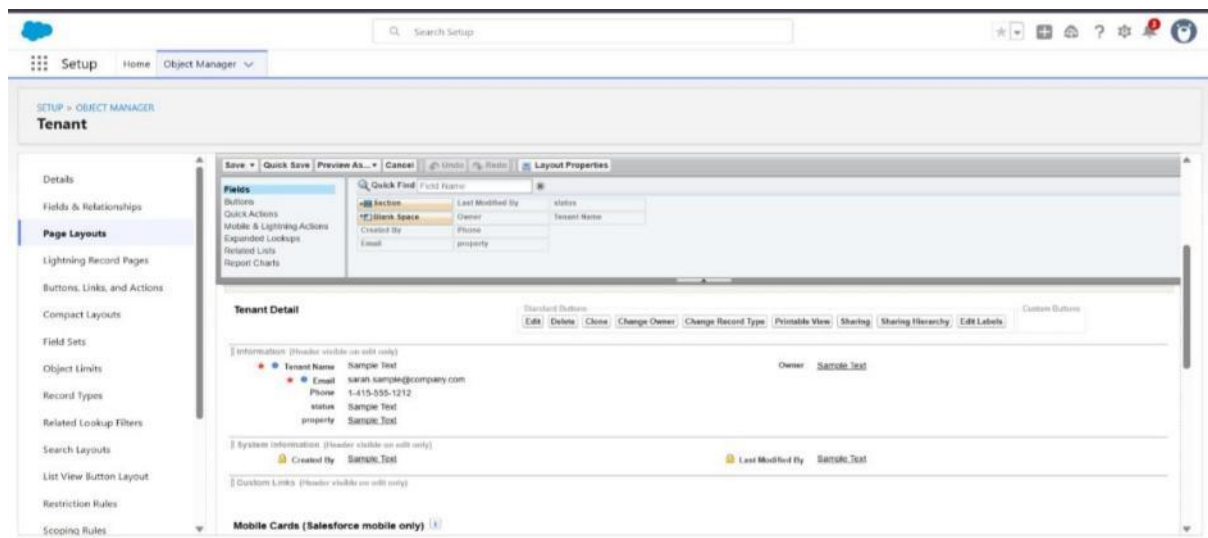
The screenshot shows the Salesforce Setup interface. In the left sidebar, 'Setup' is selected, and 'Apex Classes' is highlighted under 'Custom Code'. The main content area is titled 'Apex Classes' and 'Schedule Apex'. Below the title, there's a description: 'Schedule an Apex class that implements the Schedulable interface to be automatically executed on a specified interval.' The configuration form includes fields for 'Job Name' and 'Apex Class', both set to 'MonthlyEmailScheduler'. The 'Schedule Using' section has 'Schedule Builder' selected. The 'Schedule Apex Execution' section shows 'Frequency' set to 'Monthly', 'Start' date as '11/1/2025', 'End' date as '12/1/2025', and 'Preferred Start Time' as '9:00 AM'. There are 'Save' and 'Cancel' buttons at the top and bottom of the form.



3.11 PAGE LAYOUTS AND DYNAMIC FORMS

Page Layouts are customized for Tenant object to display only relevant fields for each user type.





This ensures the interface stays clean and relevant for users, reducing clutter and confusion.